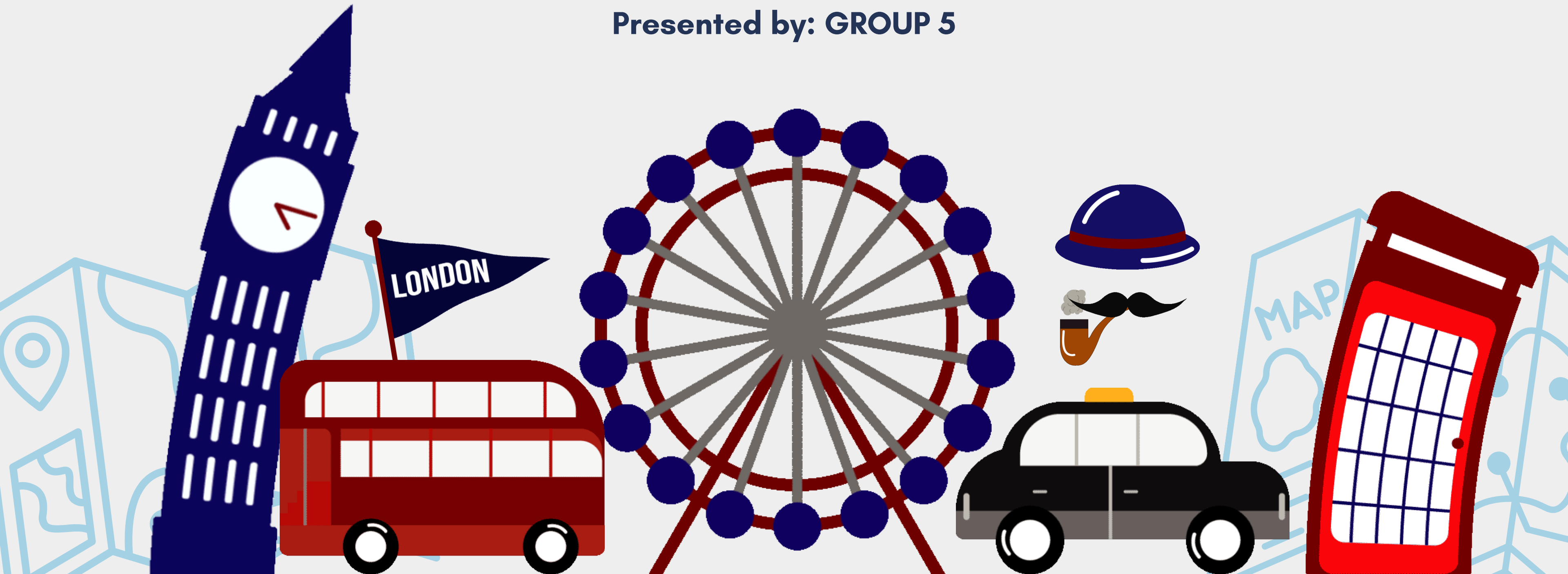


LONDON AIRBNB

DATA-230 SEC-11 DATA VISUALIZATION

Presented by: GROUP 5



MEET OUR LONDON DATA CREW



Priya



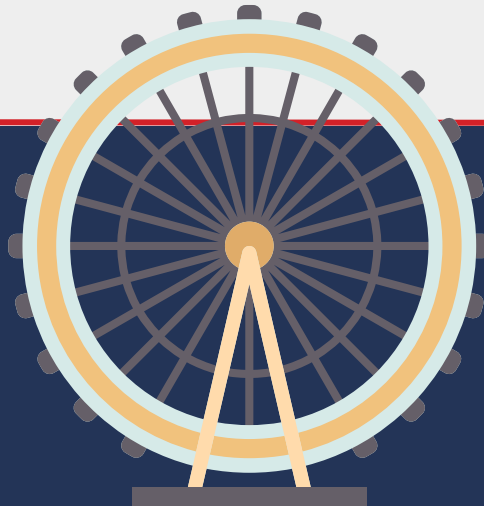
Varuna



Sanjana



Ngoc



Research Question:

WHAT MAKES A LONDON AIRBNB TOP-RATED - 4.8+?

And what can hosts learn from the listings that get it right?



Our goal: Turn 70,708 listings into actionable insights for London Airbnb hosts

Analysis journey

OUR ROADMAP TO THE ANSWER



**THE
MARKET**

What does
London Airbnb
look like?

**THE
HOST**

Does the host
behind the
listing matter?

**THE
LOCATION**

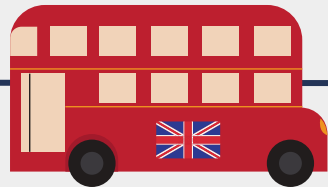
Where are
highly rated
listings
concentrated?

**THE
LISTING**

Which property
and listing
features move
together with
ratings?

**THE
ANSWER**

What should a
host focus on?

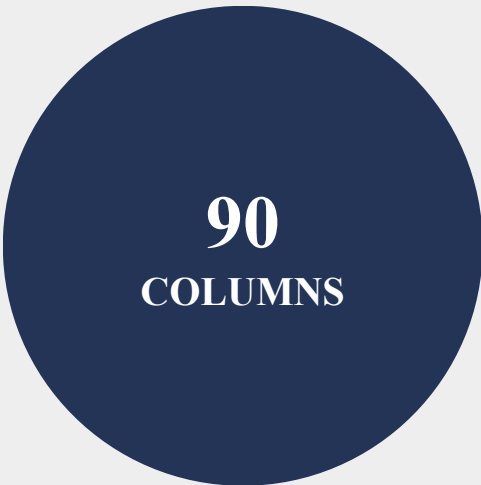


Dataset overview

92,638 LISTINGS: WHAT EARNS A 4.8+?



One row = one listing



Price, location, property, host, reviews



of 70,708 guest-rated listings scored 4.8+

SOURCE
Inside Airbnb, London
Snapshot 19 June 2026 · listings.csv.gz · prices in GBP
insideairbnb.com/get-the-data

RAW RECORDS: 3 EXAMPLE ROWS

price	bathrooms_text	bedrooms
\$127.00	1 shared bath	NaN
\$108.00	1 shared bath	1
NaN	1 bath	2

OUR TASK
Predict whether a listing is top rated (4.8+) so hosts know what to improve first.

WHY THIS DATASET
UCI Restaurant: 7,207 rows, too small
Open Food Facts: 2017, computed target
Airbnb London: 92,638 rows, 2026, guest ratings

WHY IT'S MESSY
14 empty columns
Price stored as text
33% of prices missing
Hidden placeholders
24% of listings unrated

Data cleaning

4 DECISIONS, EACH WITH A REASON

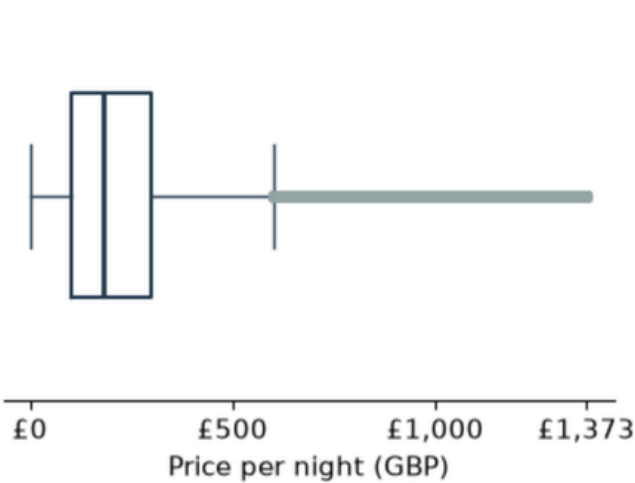
	1 DROP	2 CONVERT	3 CAP	4 FILL + FLAG
ISSUE	14 columns 100% empty	Price stored as text ("£127.00")	£527,524 price; rating 0; 2,147,483,647 nights	Missing price, beds, bedrooms, bathrooms
DECISION	Dropped	Converted to a number	Capped at £1,373 (new column); placeholders → missing	Room-type median + "missing" flag
BECAUSE	No values in this snapshot	Text cannot be charted or modelled	Entry errors and placeholders, not real values	Skewed values; missingness linked to rating

BEFORE AND AFTER

Before: listed price



After: capped at £1,373 (99th percentile)

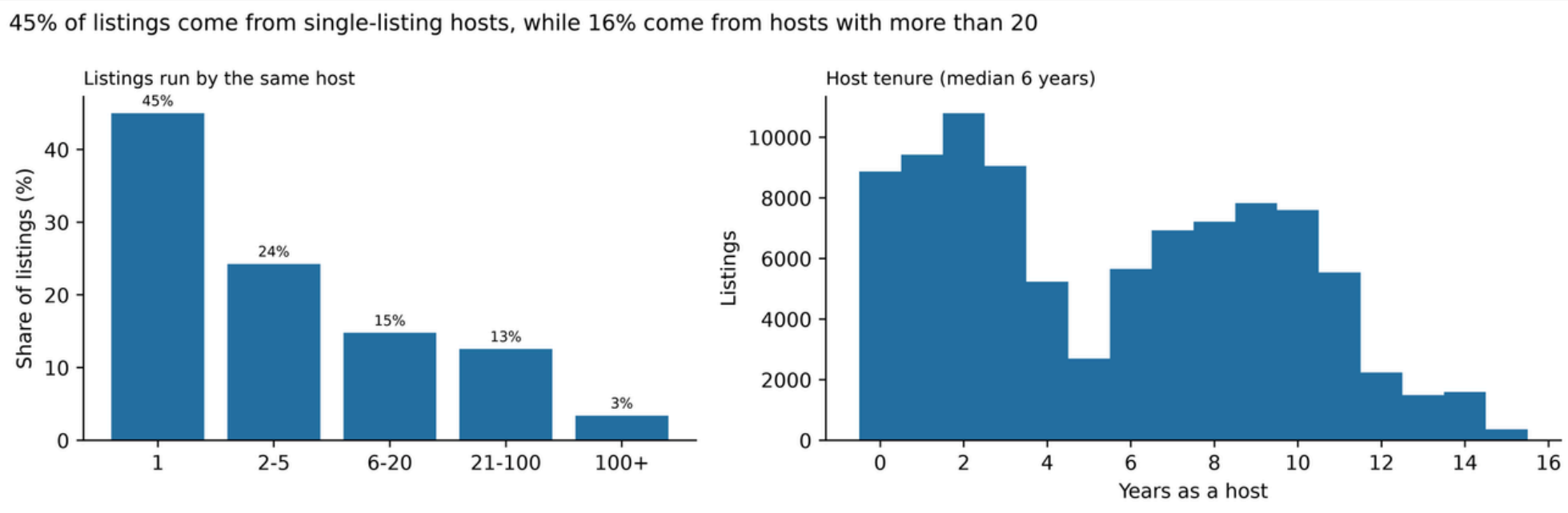


THE OUTLIER £527,524
One night, an entry error. 623 prices capped; no rated listing deleted.

CHALLENGE
33% of prices missing; 87% of those can't be booked: a pattern, not random.
Result: 92,635 × 80 · 7/7 checks pass

Univariate overview : Host Profile

WHO IS BEHIND LONDON'S AIRBNBS?



TAKEAWAY

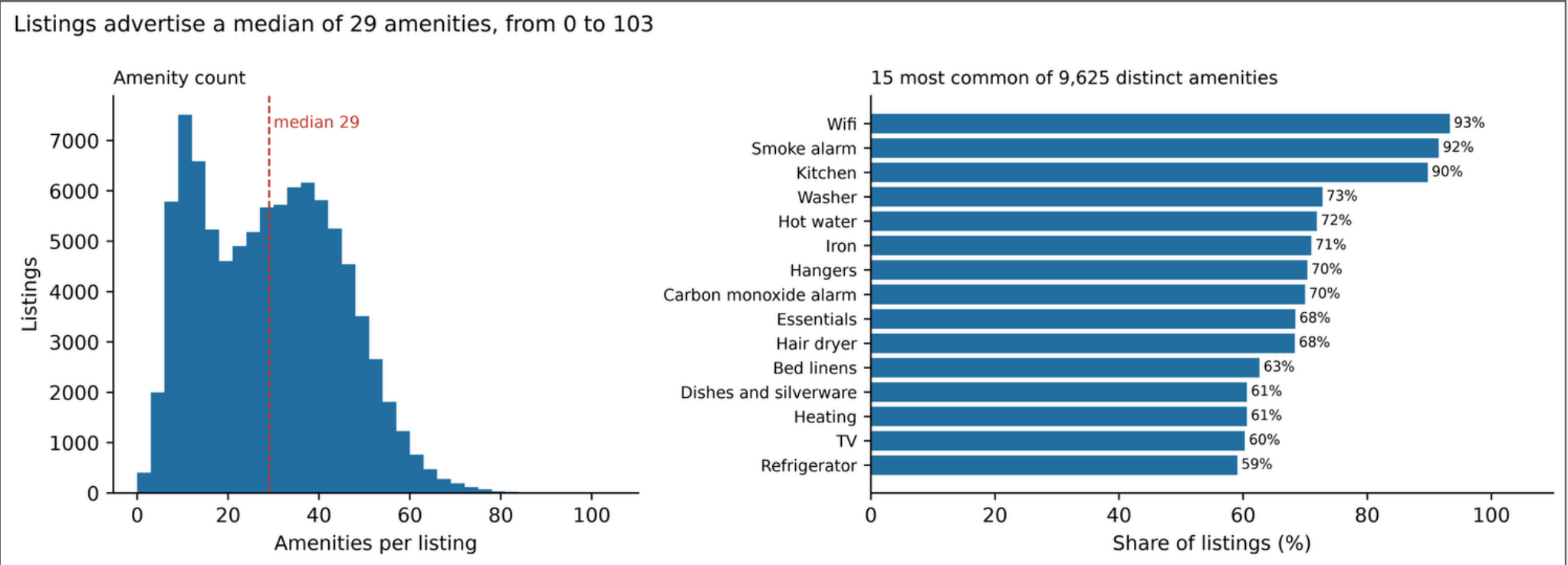
London’s host landscape is diverse: from single-listing hosts to large-scale operators.

This makes portfolio size + host tenure worth testing against ratings.

OBSERVATION	DECISION
45% from single-listing hosts; portfolios vary widely	Test portfolio size → ratings
Median tenure = 6 years	Test host tenure → ratings

Univariate overview: Listing Amenities

WHAT DO LONDON AIRBNBS OFFER?



TAKEAWAY

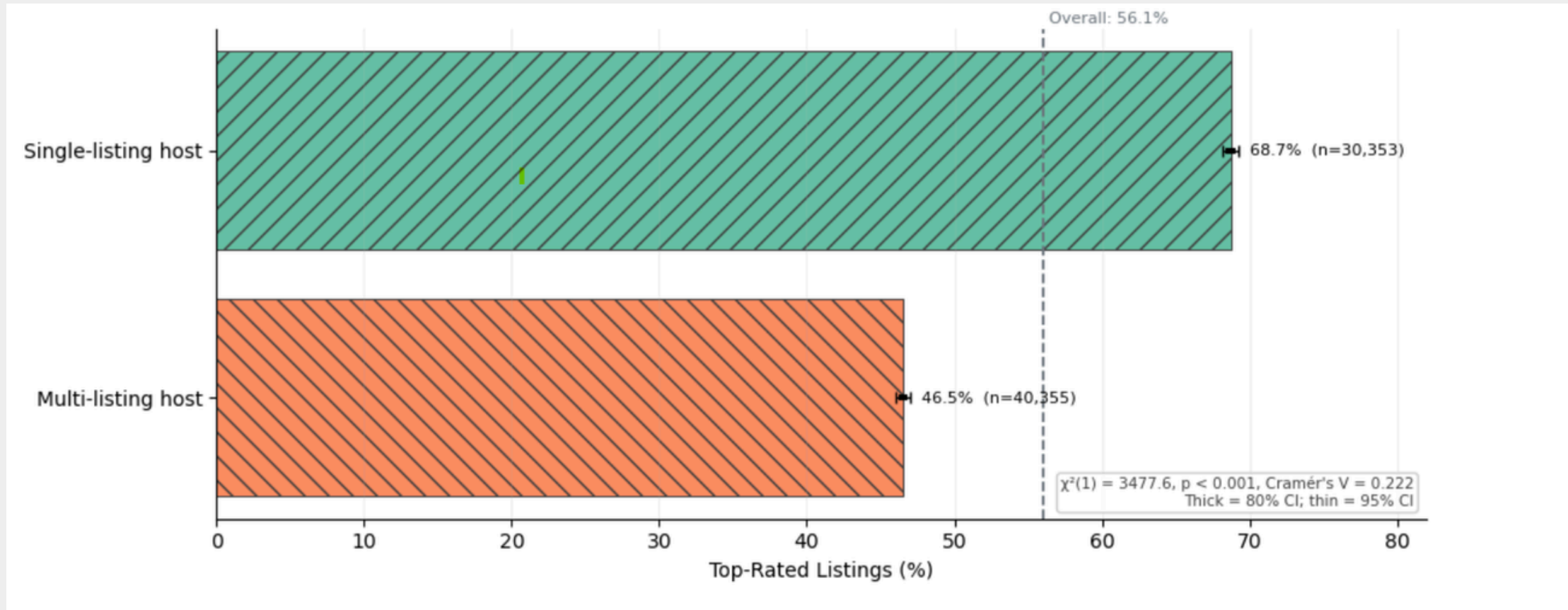
London listings vary substantially in amenity count, while basic amenities are nearly universal.

Amenity quantity and differentiating amenities move forward to rating analysis.

OBSERVATION	DECISION
Median = 29 amenities, with wide variation across listings	Test amenity count → ratings
WiFi (93%), smoke alarm (92%) & kitchen (90%) are nearly universal	Focus later analysis on amenities that differentiate listings

Bivariate overview: Host Profile

DOES HOSTING MORE LISTINGS MEAN BETTER RATINGS?



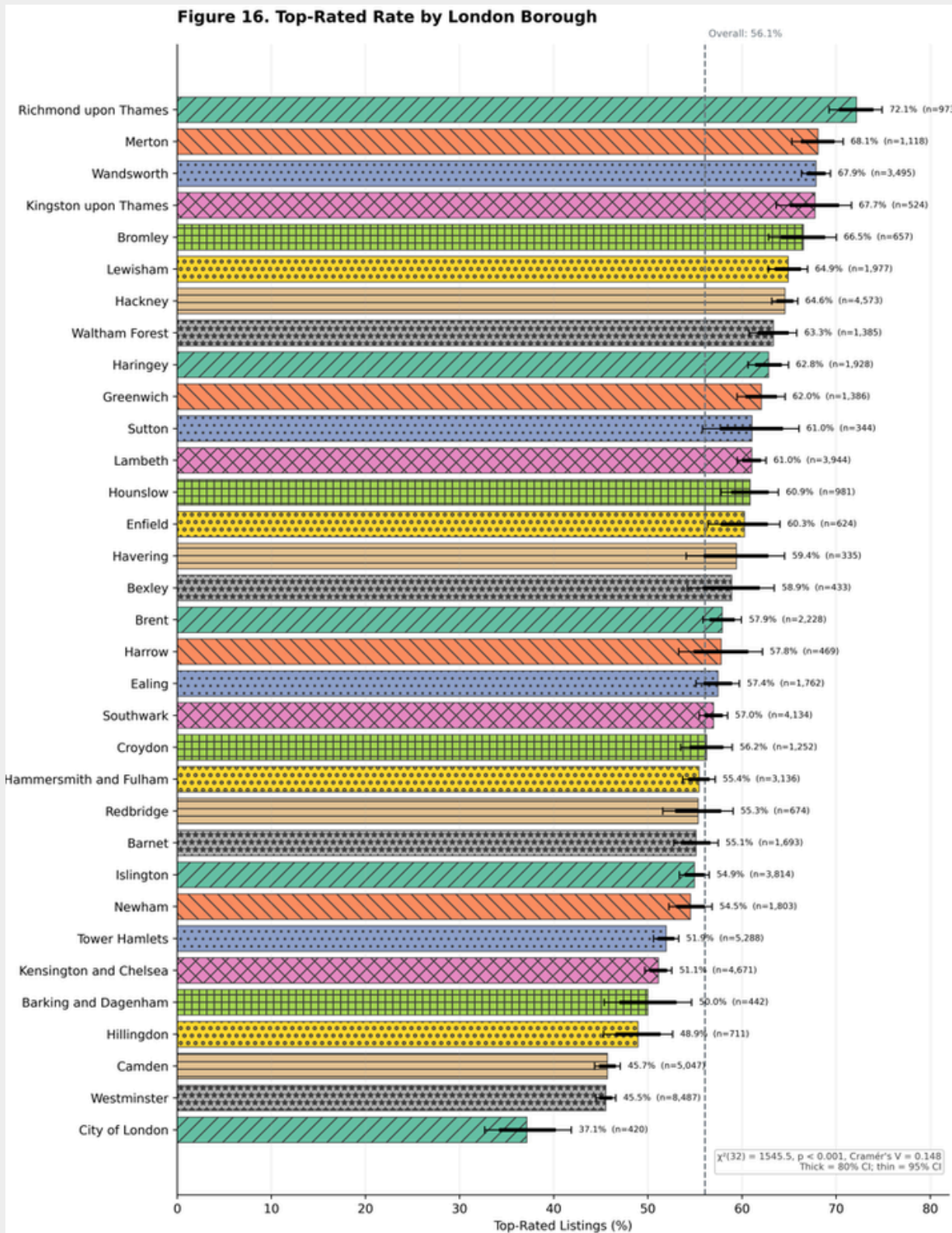
TAKEAWAY

Single-listing hosts show a 22.2 percentage-point higher top-rated rate.
Host type is our strongest categorical bivariate signal.

OBSERVATION	DECISION
Top-rated: 68.7% single-listing vs 46.5% multi-listing hosts	Retain host type as a high-priority feature
Cramér's V = 0.222, strongest categorical association examined	Carry host portfolio size into multivariate analysis

Bivariate overview: Location

DOES LOCATION INFLUENCE THE GUEST EXPERIENCE?



OBSERVATION	DECISION
Top-rated rates range from 72.1% in Richmond upon Thames to 37.1% in City of	Retain borough/location for multivariate analysis
Cramér's V = 0.148 meaningful but weaker than host type (0.222)	Treat location as a supporting predictor, not the strongest signal

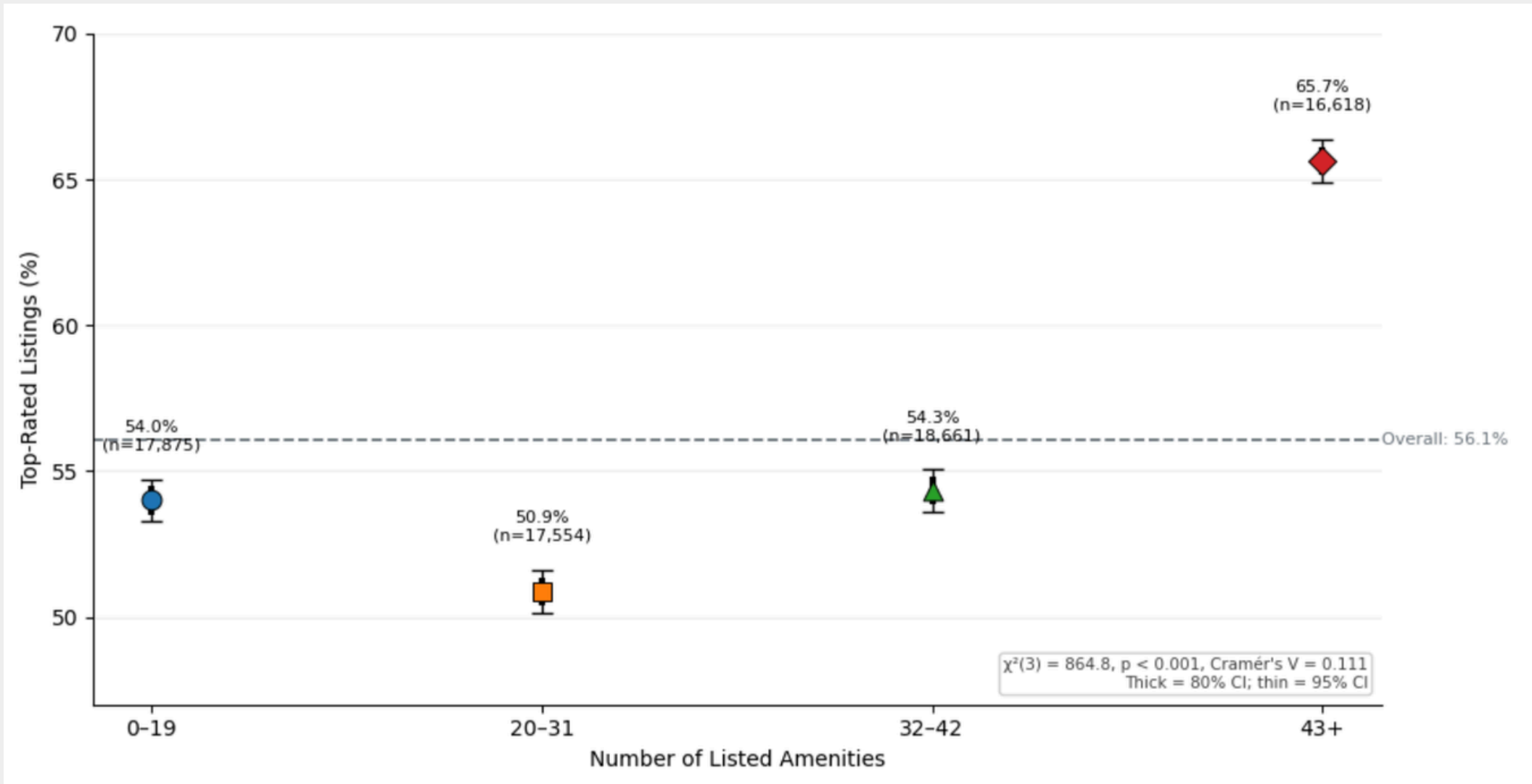
TAKEAWAY

Top-rated performance varies widely across London boroughs.

Location matters, but host type remains the stronger signal.

Bivariate overview: Amenity

WHICH LISTING FEATURES MOVE WITH RATINGS?



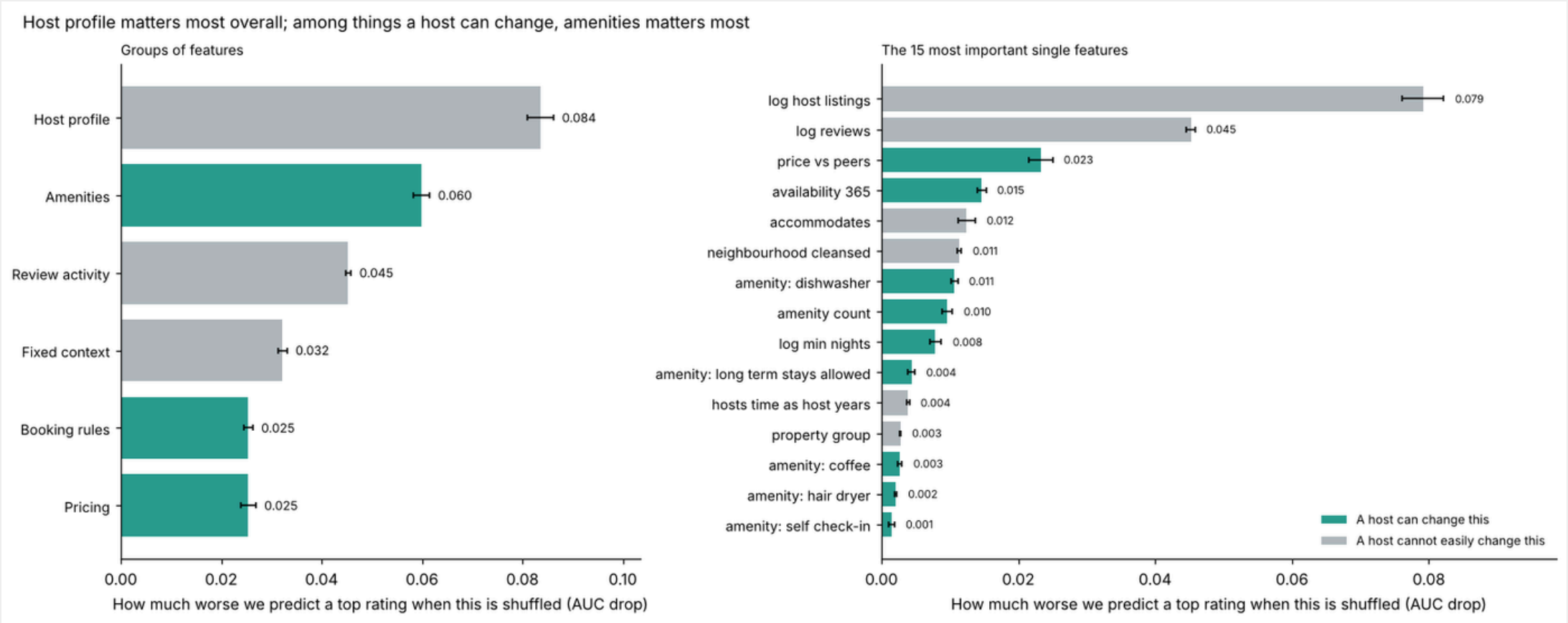
TAKEAWAY

Amenity-rich listings (43+) show the highest top-rated rate at 65.7%. Amenities matter, but their signal is weaker than host type and location.

OBSERVATION	DECISION
Listings with 43+ amenities reach 65.7% top-rated, highest among the groups	Retain amenity count for multivariate analysis
Cramér's V = 0.111 association exists, but is weaker than host type & location	Treat amenities as a supporting feature, not a dominant driver

Multivariate overview: Combined Signals

WHAT HAPPENS WHEN WE CONNECT THE SIGNALS?



TAKEAWAY

No single feature predicts a top rating (none above $p = 0.28$), but together they reach AUC 0.770.

Host profile matters most, while amenities are the strongest factor hosts can change.

OBSERVATION	DECISION
Host profile has the largest grouped importance (0.084)	Keep host profile as key context
Amenities lead host-controllable factors (0.060)	Prioritize amenities in host recommendations
Host-type gap persists in 32 of 33 boroughs	Keep host type + location together

Preliminary ML

CAN WE PREDICT WHAT DRIVES SUCCESS?

TARGET	INPUT SIGNALS	MODEL PLAN
Top-Rated $5.0 \geq 4.8$	Host profile	Logistic Regression
Not Top-Rated < 4.8	Location	Random Forest
70,708 listings	Listing + amenities	Boosted Trees
56.1% top-rated	Booking characteristics	Compare on validation set

MODELING DECISION

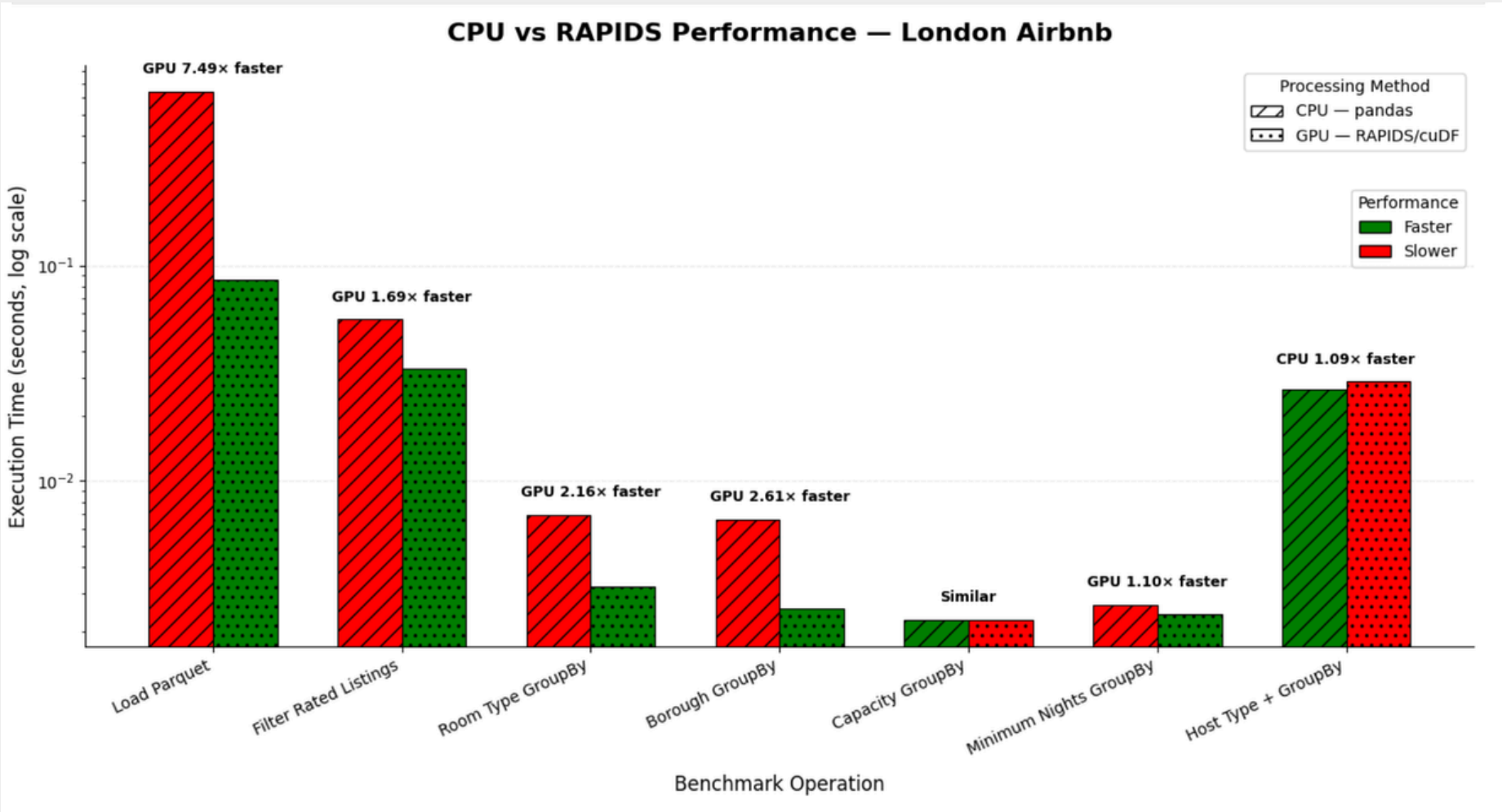
70/15/15 stratified split | Prevent target leakage | Compare performance + interpretability

TAKEAWAY

EDA identified the signals - ML tests how well they predict top-rated status together.

CPU vs GPU (RAPIDS)

DOES GPU ACCELERATION ACTUALLY HELP?



Operation	CPU (seconds)	GPU (seconds)
Data Loading (Parquet)	0.6411	0.0856
Filtering Rated Listings	0.0562	0.0332
Room Type GroupBy	0.0070	0.0032
Borough GroupBy	0.0066	0.0025
Capacity GroupBy	0.0023	0.0023
Minimum Nights GroupBy	0.0027	0.0024
Host Type + GroupBy	0.0266	0.0289

DECISION

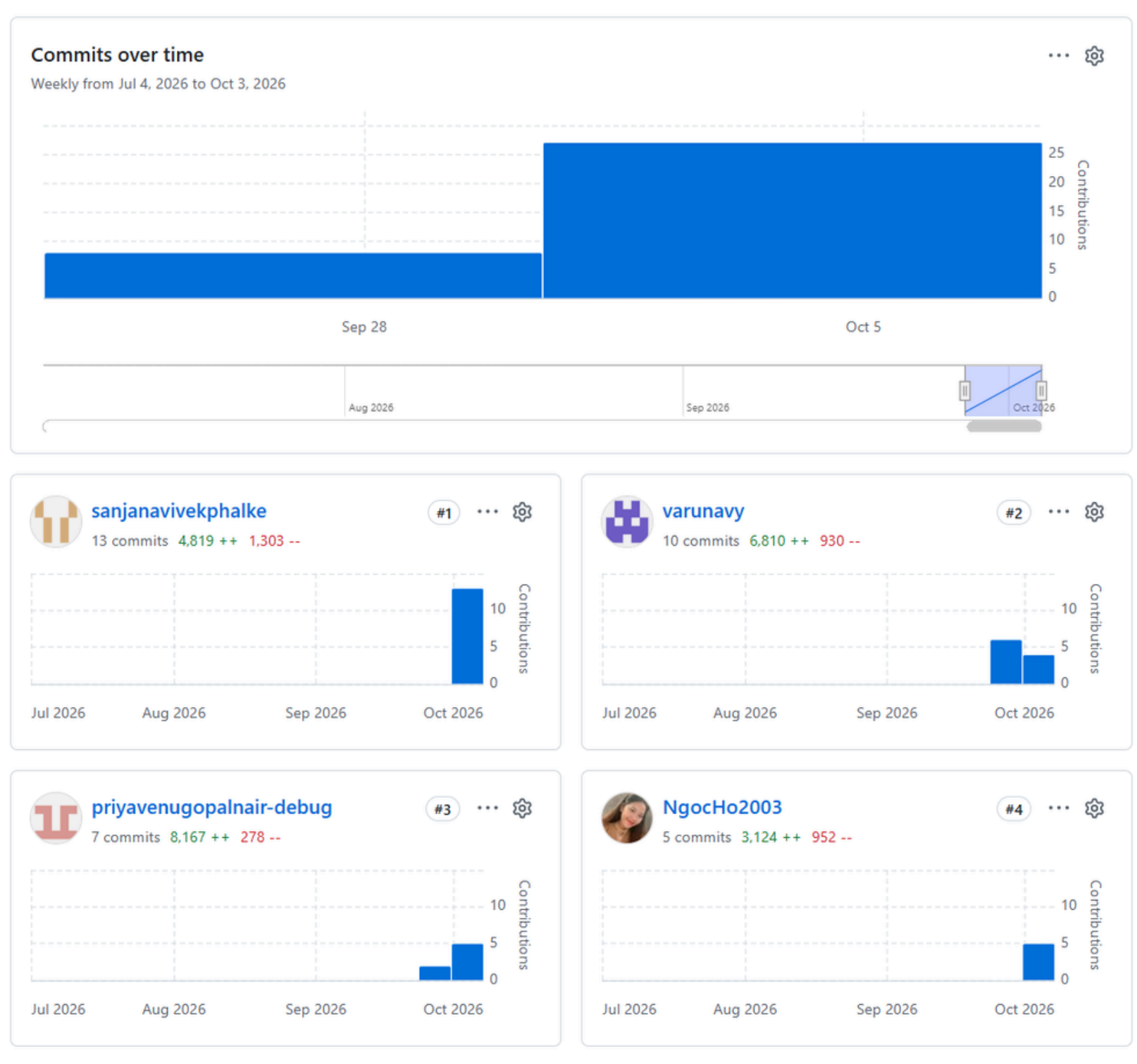
Use RAPIDS for data loading and heavier aggregations; retain pandas for lightweight operations.

TAKEAWAY

GPU ≠ always faster. RAPIDS delivers the biggest gains on heavier workloads, while pandas remains competitive on smaller operations.

CLICK TO REVIEW | GITHUB LINK | TABLEAU DASHBOARD LINK

GITHUB CONTRIBUTIONS:



Final synthesis

SO, WHAT MAKES A TOP-RATED AIRBNB?

HOST MATTERS MOST	LOCATION MATTERS	AMENITIES HELP
68.7% top-rated for single-listing hosts vs 46.5% for multi-listing hosts	Top-rated rates vary from 37.1% 72.1% across boroughs	Listings with 43+ amenities reach 65.7% top-rated
Strongest categorical signal V = 0.222	Supporting signal V = 0.148	Actionable signal V = 0.111

Top ratings are not driven by one feature. Host profile provides the strongest overall signal, while amenities are the strongest factor hosts can directly change.

Recommendations

THE HOST PLAYBOOK

STRENGTHEN THE STAY	OPTIMIZE THE OFFER	COMPETE SMARTER
Prioritize useful, differentiating amenities not just the basics	Review pricing and booking rules alongside comparable listings	Benchmark against similar hosts and the same borough
Why? Amenities are the strongest host-controllable feature group (0.060)	Why? Pricing & booking rules remain actionable signals (0.025 each)	Why? Host profile (0.084) and location influence rating patterns

FINAL MESSAGE

Hosts cannot change their location or history but they can improve how their listing competes within that context.



70,708 London Airbnbs
One question. One clear takeaway.

**Top-rated success comes from a
combination of context and choices,
amenities are the strongest lever hosts
can directly control**

THANK YOU